

ELEVATE LABS

TASK 11

Phishing Attack Simulation & Detection

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Domain: Cybersecurity

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1. Understand phishing attacks

Phishing attacks are a form of social engineering in which attackers trick users into revealing sensitive information by pretending to be a trusted source such as a bank, organization, or service provider. These attacks are commonly carried out through emails, messages, phone calls, or fake websites that create a sense of urgency or fear, prompting users to act quickly without verification. Phishing does not rely on technical hacking alone; instead, it exploits human trust and behavior, making it one of the most effective and widely used cyberattacks.

Types:-

Email phishing is the most common type, where fraudulent emails contain malicious links or attachments

Spear phishing targets specific individuals or organizations using personalized information

Whaling is a high-level spear phishing attack aimed at executives or senior officials.

Smishing uses SMS or messaging apps to send fake alerts or links

vishing involves phone calls where attackers pose as bank or support staff

Clone phishing copies a legitimate email but replaces the original link or attachment with a malicious one.

2. Create fake email template

Creating a fake email template in GoPhish is done for security awareness training and phishing simulation, not for real attacks. The purpose is to help organizations understand how phishing emails look, how users react to them, and how to improve employee awareness. In a phishing simulation, a realistic-looking email template is designed to test whether users can identify suspicious messages. These templates mimic common phishing characteristics such as urgent language, branding style, and call-to-action buttons, but are used only in controlled environments with prior authorization. The results help security teams measure risk, improve training, and strengthen email security policies.

Step 1: Download GoPhish

Wget

<https://github.com/gophish/gophish/releases/download/v0.12.1/gophish-v0.12.1-linux-64bit.zip>

```
(kali@kali)~$ wget https://github.com/gophish/gophish/releases/download/v0.12.1/gophish-v0.12.1-linux-64bit.zip
--2026-02-02 07:24:23-- https://github.com/gophish/gophish/releases/download/v0.12.1/gophish-v0.12.1-linux-64bit.zip
Resolving github.com (github.com)... 20.207.73.82
Connecting to github.com (github.com)|20.207.73.82|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/14508450/d775c2cd-8d50-49e6-a442-fec1d3d0f1a3?sp=r&sv=2018-11-09&sr=b&spr=https&se=2026-02-02T13K3A10K
[2.1.1-linux-64bit.zip]
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.109.133, 185.199.108.133, 185.199.111.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)|185.199.109.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 33327057 (32M) [application/octet-stream]
Saving to: 'gophish-v0.12.1-linux-64bit.zip'

gophish-v0.12.1-linux-64bit.zip 100%[=====]
2026-02-02 07:24:29 (6.18 MB/s) - 'gophish-v0.12.1-linux-64bit.zip' saved [33327057/33327057]

(kali@kali)~$
```

Extract files:

unzip gophish-v0.12.1-linux-64bit.zip

cd gophish

```
(kali@kali)-[~]
$ unzip gophish-v0.12.1-linux-64bit.zip
cd gophish

Archive: gophish-v0.12.1-linux-64bit.zip
  inflating: gophish
    creating: static/js/dist/
    creating: static/js/dist/app/
  inflating: static/js/dist/app/sending_profiles.min.js
  inflating: static/js/dist/app/campaign_results.min.js
  inflating: static/js/dist/app/gophish.min.js
  inflating: static/js/dist/app/campaigns.min.js
  inflating: static/js/dist/app/autocomplete.min.js
  inflating: static/js/dist/app/settings.min.js
  inflating: static/js/dist/app/users.min.js
  inflating: static/js/dist/app/webhooks.min.js
  inflating: static/js/dist/app/dashboard.min.js
  inflating: static/js/dist/app/passwords.min.js
  inflating: static/js/dist/app/templates.min.js
  inflating: static/js/dist/app/groups.min.js
  inflating: static/js/dist/app/landing_pages.min.js
  inflating: static/js/dist/vendor.min.js
    creating: static/js/src/vendor/ckeditor/
  inflating: static/js/src/vendor/ckeditor/CHANGES.md
    creating: static/js/src/vendor/ckeditor/skins/
    creating: static/js/src/vendor/ckeditor/skins/moono-lisa/
  inflating: static/js/src/vendor/ckeditor/skins/moono-lisa/editor_ie.css
```

Give execution permission:

chmod +x gophish

```
(kali@kali)-[~]
$ chmod +x gophish
```

Step 2: Start GoPhish Server

Run GoPhish:

sudo ./gophish

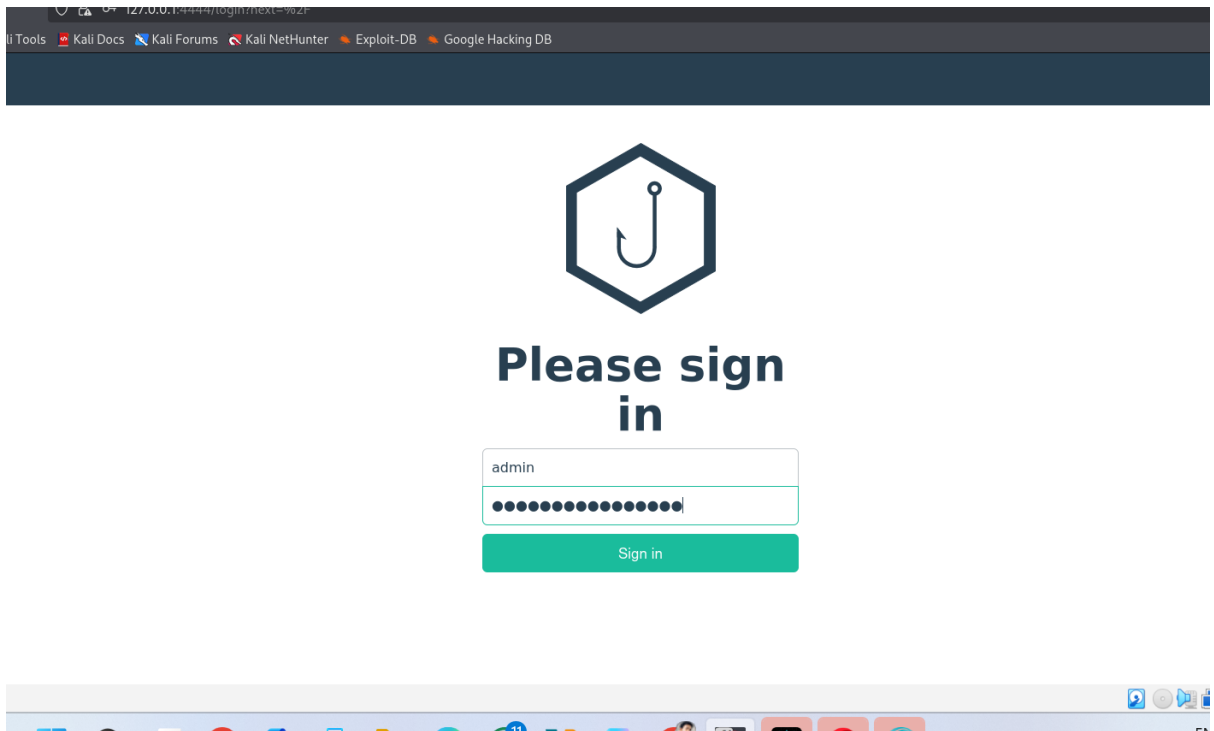
```
(kali@kali)-[~]
$ sudo ./gophish

time="2026-02-02T07:50:52-05:00" level=warning msg="No contact address has been configured."
time="2026-02-02T07:50:52-05:00" level=warning msg="Please consider adding a contact_address entry in your config.json"
goose: migrating db environment 'production', current version: 0, target: 20220321133237
OK      20160118194630_init.sql
OK      20160131153104_0.1.2_add_event_details.sql
OK      20160211211220_0.1.2_add_ignore_cert_errors.sql
OK      20160217211342_0.1.2_create_from_col_results.sql
OK      20160225173824_0.1.2_capture_credentials.sql
OK      20160227180335_0.1.2_store-smtp-settings.sql
OK      20160317214457_0.2_redirect_url.sql
OK      20160605210903_0.2_campaign_scheduling.sql
OK      20170104220731_0.2_result_statuses.sql
OK      20170219122503_0.2.1_email_headers.sql
OK      20170827141312_0.4_utc_dates.sql
OK      20171027213457_0.4.1_maillogs.sql
OK      20171208201932_0.4.1_next_send_date.sql
```

Step 3: Access GoPhish Dashboard

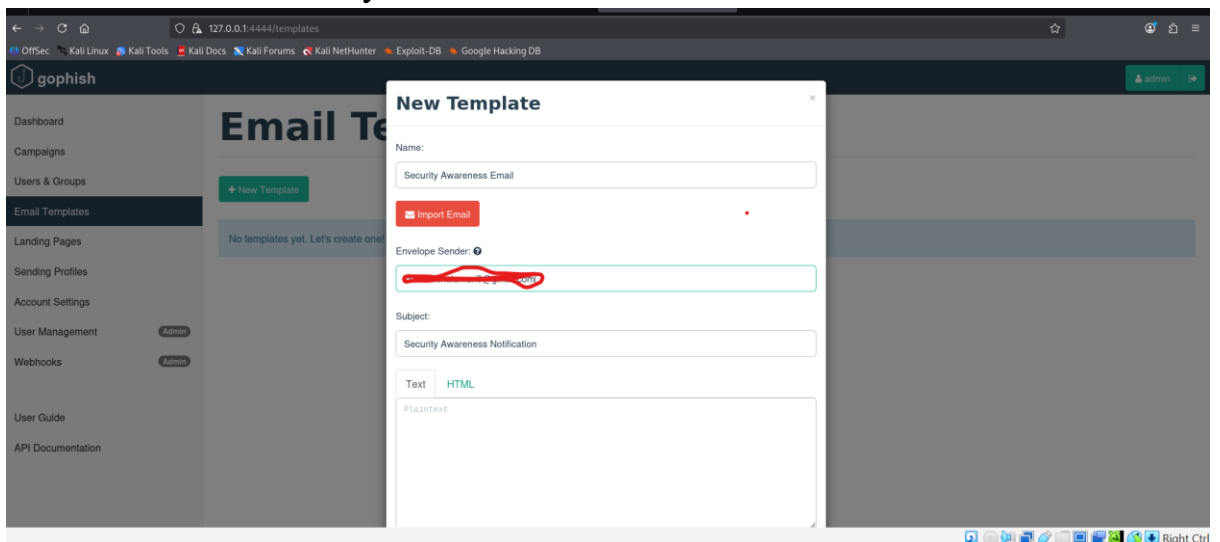
Open browser and go to:

<https://127.0.0.1:3333>



Step 4: Create Email Template

1. Dashboard → **Email Templates**
2. Click **New Template**
3. Enter details:
 - **Name:** Security Awareness Email
 - **Subject:** Security Awareness Notification
 - **From:** IT Security Team



Step 5: Add Email Body (Training Purpose)

gophish

Dashboard

Campaigns

Users & Groups

Email Templates

Landing Pages

Sending Profiles

Account Settings

User Management Admin

Webhooks Admin

User Guide

API Documentation

Email Template

+ New Template

No templates yet. Let's create one

Import Email

Envelope Sender:

Subject:

Text ☒ HTML

Hello {{.xxxxxx}},

This email is part of a security awareness exercise.
Please review this message carefully.

Click the link below to continue:
{{.stthomasedu-ucor-us/microsoft.html}}

Thank you,
IT Security Team

☒ Add Tracking Image

+ Add Files

Show entries

Search:

Name

Step 6: Save Template

- Click Save Template
- Email template is successfully created

gophish

Dashboard

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Landing Pages

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Account Settings

User Management Admin

Webhooks Admin

User Guide

API Documentation

Email Templates

+ New Template

Template added successfully!

Show entries

Search:

Name	Modified Date
Security Awareness Email	February 2nd 2026, 8:23:42 am

Showing 1 to 1 of 1 entries

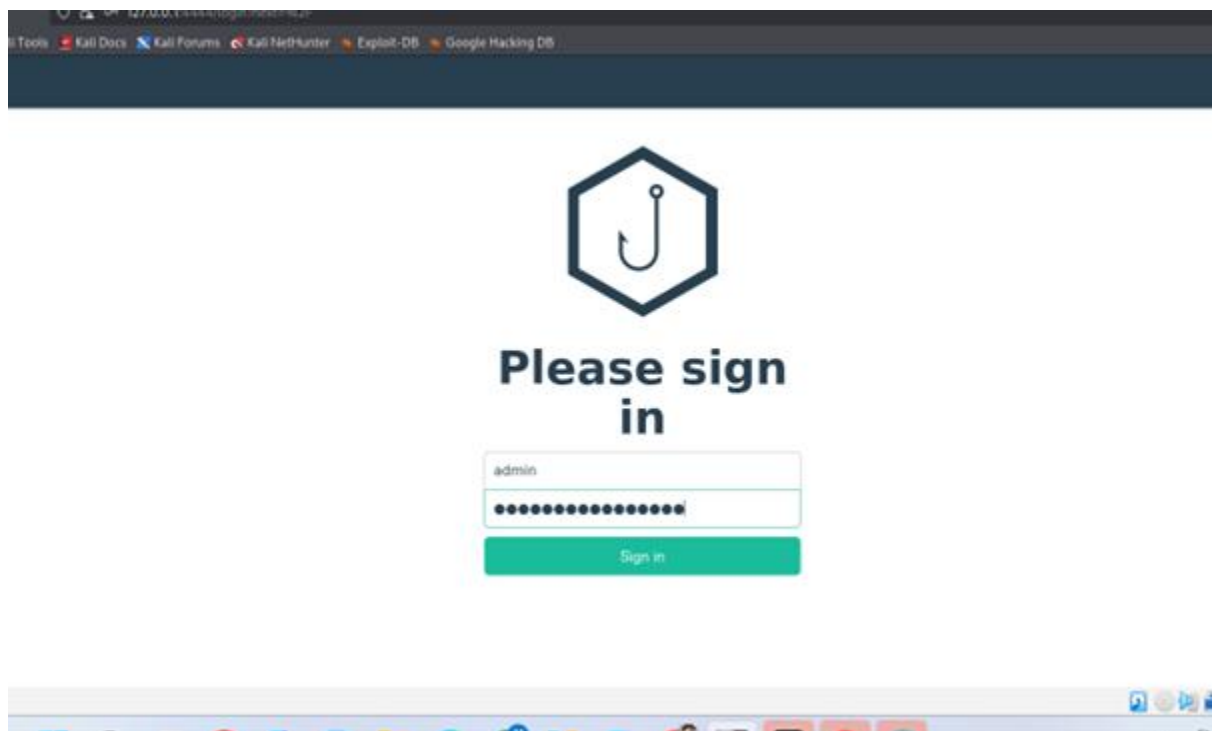
Previous 1 Next

3. Setup Landing Page

A landing page in **GoPhish** is the web page that users are redirected to after clicking link in a phishing simulation email. In security awareness training, landing pages are used to **educate users**, record click behavior, and demonstrate how phishing attacks attempt to collect information. Instead of stealing real credentials, an ethical landing page displays a warning or training message explaining that the user has clicked a simulated phishing link. Setting up a landing page helps organizations analyze user awareness, improve training programs, and reduce the risk of real phishing attacks.

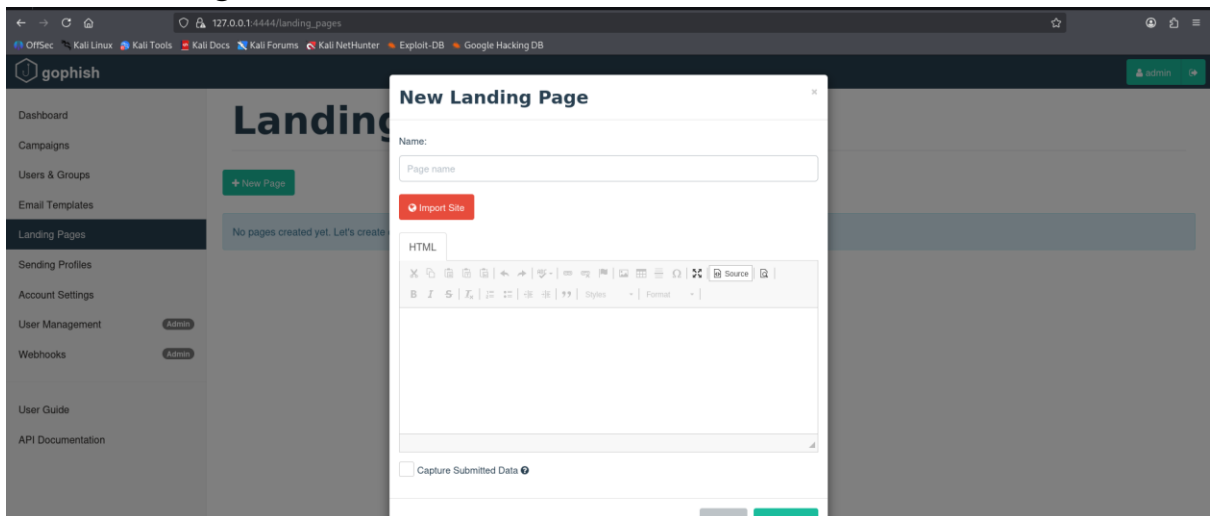
Step 1: Open GoPhish Dashboard

- Start GoPhish
- Open browser
- Go to:



Step 2: Navigate to Landing Pages

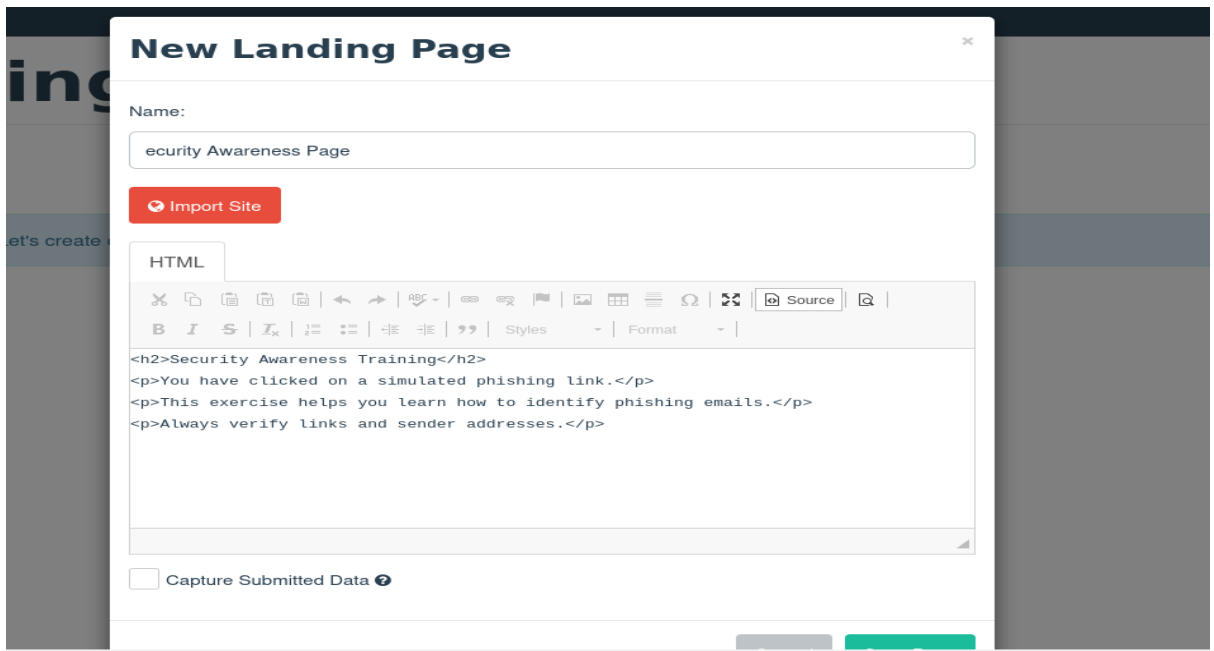
- Dashboard → Landing Pages
- Click New Page



Step 3: Create Landing Page Details

Fill the following fields:

- Name: Security Awareness Page
- HTML Content:



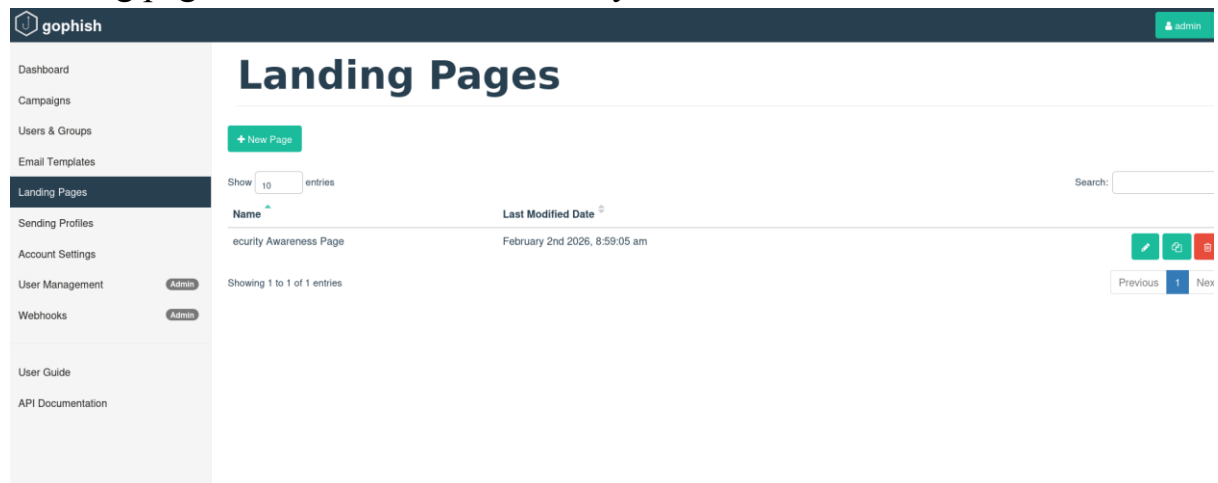
Step 4: Configure Page Settings

Enable / Disable options carefully:

- Capture Submitted Data: **OFF**
- Capture Passwords: **OFF**
- Redirect to URL:

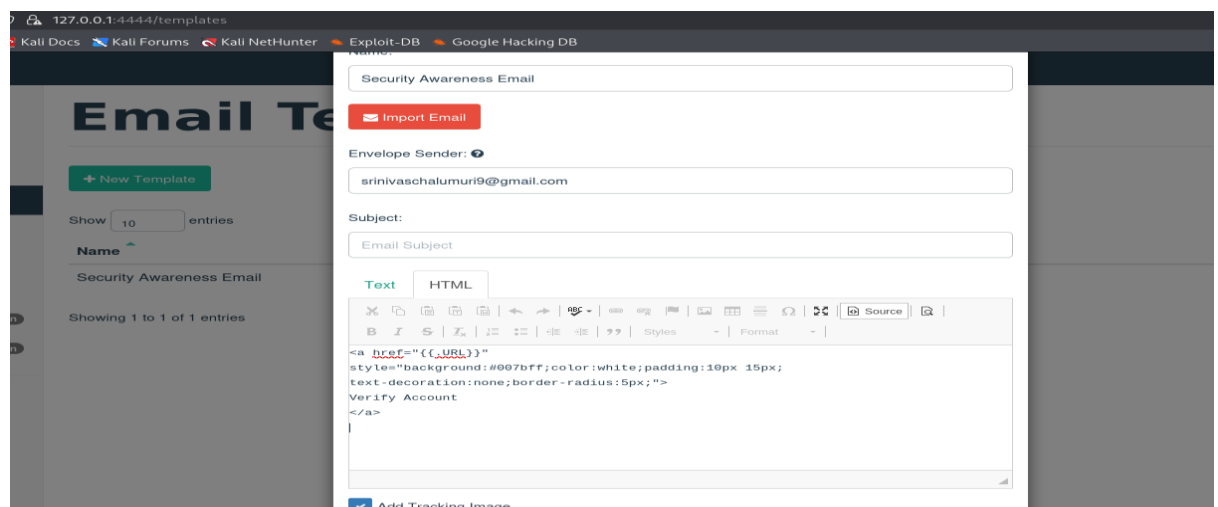
Step 5: Save Landing Page

- Click **Save Page**
- Landing page is now created successfully



Step 6: Link Landing Page to Email Template

- Go to **Email Templates**
- Edit your template
- Use:

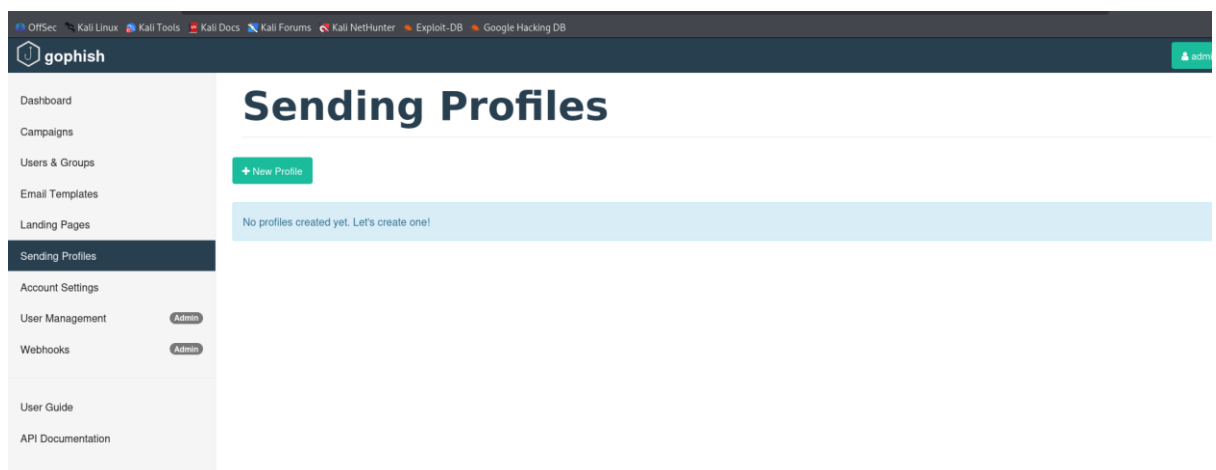


4. Send test phishing email

Sending a test phishing email in GoPhish is part of a controlled security awareness simulation. The purpose is to verify that the email template, landing page, and sending profile are correctly configured before launching a full campaign. Test emails help security teams ensure that links work properly, emails are delivered successfully, and tracking is enabled. This step reduces errors and ensures the phishing simulation is conducted safely, ethically, and with authorization.

Step 1: Create Sending Profile (If Not Created)

1. GoPhish Dashboard → **Sending Profiles**
2. Click **New Profile**
3. Fill details:
 - **Name:** Test SMTP Profile
 - **From:** IT Security itsecurity@test.com
 - **Host:** smtp.gmail.com:587 (example)
 - **Username:** sender email
 - **Password:** app password
4. Enable TLS
5. Click **Save Profile**



New Sending Profile

Name:
Test SMTP profile

Interface Type:
SMTP

SMTP From: ⓘ
IT Security itsecurity@test.com

Host:
smtp.gmail.com:587

Username:
snu

Password:
●●●●●●●●

☒ Ignore Certificate Errors ⓘ

Email Headers:

5. Track responses.

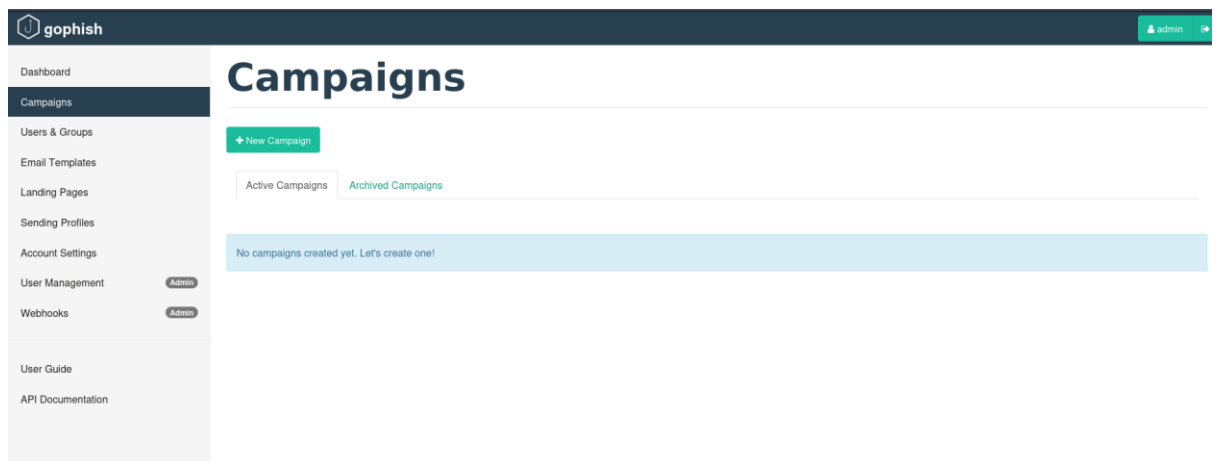
Tracking responses in GoPhish is used to monitor how users interact with a phishing awareness campaign. It helps security teams understand whether users open emails, click links, or submit data. By analyzing these responses, organizations can measure user awareness, identify risky behavior, and improve security training. Response tracking in GoPhish provides real-time statistics and detailed reports for evaluating the effectiveness of phishing simulations.

Step 1: Open Campaign Dashboard

Open GoPhish dashboard

Click on Campaigns

Select the running or completed campaign



Step 2: View Campaign Statistics

On the campaign page, observe the following metrics:

Emails Sent – total number of emails delivered

Emails Opened – number of users who opened the email

Clicked Links – users who clicked the phishing link

Submitted Data – users who entered information

Reported – users who reported the email (if enabled)

Details

Show entries

Search:

First Name	Last Name	Email	Position	Status	Reported
sr	ch	srinivaschalumuri9@gmail.com		Error	☐
sr	vd	srinivaschalumuri491@gmail.com		Error	☐
srin	vas	sinivaschalumuri9@gmail.com		Error	☐

Showing 1 to 3 of 3 entries

Previous 1

Step 3: Timeline Analysis

View the timeline graph

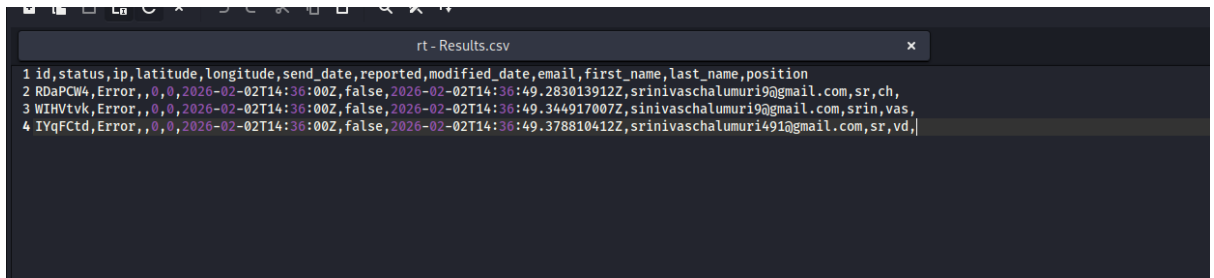
Analyze when users interacted with the email

Identify quick clicks indicating high risk

Step 5: Export Results

Click Export Results (CSV)

Save the report for documentation or lab record



```
rt - Results.csv
1 id,status,ip,latitude,longitude,send_date,reported,modified_date,email,first_name,last_name,position
2 RDaPCW4,Error,,0,0,2026-02-02T14:36:00Z,false,2026-02-02T14:36:49.283013912Z,srinivaschalumuri19@gmail.com,sr,ch,
3 WIHvtvk,Error,,0,0,2026-02-02T14:36:00Z,false,2026-02-02T14:36:49.344917007Z,sinivaschalumuri19@gmail.com,srin,vas,
4 IYqFctd,Error,,0,0,2026-02-02T14:36:00Z,false,2026-02-02T14:36:49.378810412Z,srinivaschalumuri491@gmail.com,sr,vd,
```

6. Identify Red Flags

Red flags in phishing emails are warning signs that indicate a message may be fraudulent or malicious. Identifying these red flags helps users avoid falling victim to phishing attacks. In GoPhish awareness campaigns, red flags are intentionally included in email templates so users can learn how attackers manipulate trust, urgency, and curiosity. Recognizing red flags improves user awareness and reduces the success rate of real phishing attacks.

7. Learn Prevention

Phishing prevention refers to the methods and practices used to protect users and organizations from phishing attacks. Since phishing targets human behavior rather than system vulnerabilities, prevention focuses on user awareness, secure email practices, and technical controls. Learning prevention helps reduce the success rate of phishing attacks and minimizes data breaches, financial loss, and identity theft.

8. Document Simulation

- GoPhish is used for phishing awareness and simulation
- Email template is created to simulate phishing emails
- Landing page is set to capture user interaction safely
- SMTP sending profile is configured for email delivery
- User group is created with test email addresses
- Campaign is created using template and landing page
- Test phishing email is sent to users
- User actions like open and click are tracked
- Phishing red flags are identified from the email
- Results are analyzed to improve user awareness